Università degli studi di Padova

Stochastic Methods for Finance

Report 4: Value at Risk of a two-asset Portfolio

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1 Introduction

This report estimates the Value at Risk (VaR) of a two-asset portfolio and evaluates the diversification effect, defined as the reduction in risk obtained by combining assets with imperfectly correlated returns.

The analysis considers ExxonMobil (XOM) and Microsoft (MSFT), two firms with different sector exposures and risk profiles. XOM operates in the energy sector and exhibits relatively stable behavior (beta = 0.29), while MSFT belongs to the technology sector and is characterized by higher growth exposure and volatility (beta = 1.10).

Their combination is suitable for studying diversification, as imperfect correlation reduces the probability of joint extreme losses and allows for a comparison of risk at both asset and portfolio level.

2 Parametric Normal VaR – sigma flat

2.1 Data and Methodology

We use daily price data over a six-month period (Refinitiv, as of April 2, 2026) for XOM and MSFT. An equally weighted portfolio is formed on October 2, 2025, with an initial value of \$100,000 and daily rebalancing.

Daily percentage returns are used to estimate mean, variance, and volatility for each asset and for the portfolio:

- the average daily return (μ)
- the variance (σ^2)
- the standard deviation (volatility, σ)

	XOM	MSFT	portfolio
μ	0.31%	-0.25%	0.03%
σ^2	0.02%	0.03%	0.01%
vol, σ	1.54%	1.69%	1.03%

Portfolio returns are computed as a weighted average of asset returns. Portfolio volatility accounts for correlation between assets, estimated at $\rho = -0.18$, which reduces overall risk.

VaR is computed under the assumption of normally distributed returns and constant volatility. The 1-day VaR at confidence level X is:

$$\mathrm{VaR}(X) = -\mu + z_X \sigma \tag{1}$$

where z_X is the standard normal quantile. Monetary VaR is obtained by scaling with portfolio value. For horizon T, assuming i.i.d. returns over time, VaR is scaled using the square-root-of-time rule:

$$\mathrm{VaR}_T = -\mu T + z_X \sigma \sqrt{T} \tag{2}$$

Negative correlation between assets reduces portfolio volatility below individual levels, generating diversification benefits. This is measured as:

$$\text{Diversification Benefit} = \sum \mathrm{VaR}_i - \mathrm{VaR}_{\text{portfolio}} \tag{3}$$

2.2 Results and Discussion

	XOM VaR ($)			MSFT VaR ($)			Portfolio VaR ($)			Div. benefit ($)		
T	95%	99%	99.5%	95%	99%	99.5%	95%	99%	99.5%	95%	99%	99.5%
1	1136	1680	1877	1553	2151	2368	1710	2441	2706	979	1391	1540
5	2101	3318	3759	3834	5172	5657	3746	5380	5973	2189	3110	3443
10	2506	4228	4852	5805	7697	8383	5215	7527	8365	3096	4398	4870
50	1218	5068	6463	16592	20824	22357	10888	16058	17931	6922	9834	10889
100	-2925	2520	4493	27293	33278	35446	14579	21890	24539	9789	13907	15400

Table 1: Value at Risk and Diversification Benefit across horizons – sigma flat

Table 1 shows that VaR increases with confidence level and horizon for all assets. Portfolio VaR remains below the sum of individual VaRs, confirming diversification gains, which become more pronounced at longer horizons. Negative VaR may occur for long horizons at low confidence levels when the expected return component dominates volatility, shifting the lower tail of the return distribution above zero. This result reflects the assumptions of the normal i.i.d. framework rather than a true absence of downside risk.

The model assumes normality and constant volatility, which is restrictive given volatility clustering in financial returns. The next section introduces a time-varying volatility framework using the RiskMetrics EWMA approach.

3 Parametric Normal VaR – EWMA Volatility

3.1 Methodology

Volatility is estimated using the RiskMetrics Exponentially Weighted Moving Average (EWMA) model, which allows for time-varying conditional variance.

The variance evolves according to:

$$\sigma_t^2 = \lambda\sigma_{t-1}^2 + (1-\lambda)r_{t-1}^2 \tag{4}$$

where $\lambda = 0.94$ is the decay factor and r_{t-1} denotes the return at time $t-1$.

	XOM	MSFT	portfolio
μ	0.31%	-0.25%	0.03%
σ^2	0.03%	0.03%	0.01%
vol, σ	1.62%	1.78%	0.88%

This specification assigns higher weight to recent observations, capturing volatility clustering in financial returns.

3.2 Results and Discussion

EWMA volatility is generally higher for individual assets, reflecting increased sensitivity to recent market movements. Despite this, portfolio volatility remains lower due to diversification effects driven by imperfect correlation.

	XOM VaR ($)			MSFT VaR ($)			Portfolio VaR ($)			Div. benefit ($)		
T	95%	99%	99.5%	95%	99%	99.5%	95%	99%	99.5%	95%	99%	99.5%
1	1209	1785	1993	1634	2267	2496	1456	2080	2306	1388	1972	2183
5	2266	3553	4019	4016	5430	5943	3178	4574	5080	3103	4409	4882
10	2739	4559	5219	6062	8062	8787	4412	6386	7102	4389	6235	6904
50	1739	5809	7284	17167	21641	23262	9093	13507	15107	9814	13942	15438
100	-2188	3568	5653	28106	34433	36725	12040	18283	20545	13878	19718	21833

Table 2: Value at Risk and Diversification Benefit across horizons – EWMA

Accordingly, portfolio VaR increases relative to the flat-volatility case but remains below the sum of individual VaRs, confirming persistent diversification benefits under time-varying volatility.

4 Monte Carlo Value at Risk

4.1 Methodology

The Monte Carlo approach estimates VaR by simulating a large number of possible future return scenarios and evaluating the resulting portfolio outcomes.

Starting from the current portfolio value, correlated returns are generated from a multivariate normal distribution and used to revalue the portfolio. This process is repeated N times to approximate the full profit and loss distribution.

Correlation between assets is explicitly preserved in the simulation to ensure realistic joint behavior.

VaR is computed as the empirical quantile of the simulated loss distribution:

$$\mathrm{VaR}_X = -\mathrm{Quantile}_{1-X}(\Delta P) \tag{5}$$

where ΔP denotes simulated profit and loss.

Confidence levels of 95%, 99%, and 99.5% are considered. For horizon T, time scaling is implemented by adjusting the mean linearly in T and volatility with $\sqrt{T}$, consistent with the i.i.d. normal assumption.

4.2 Results and Discussion

The Monte Carlo results are broadly consistent with the parametric framework, reflecting the assumption of normally distributed returns.

Portfolio VaR remains below the sum of individual VaRs across all horizons, confirming diversification benefits. In addition, the simulation framework provides greater flexibility and can be extended to non-normal distributions or more complex dependence structures.

	XOM VaR ($)			MSFT VaR ($)			Portfolio VaR ($)			Div. benefit ($)		
T	95%	99%	99.5%	95%	99%	99.5%	95%	99%	99.5%	95%	99%	99.5%
1	1152	1698	1841	1547	2124	2337	1752	2414	2642	948	1409	1535
5	2085	3374	3631	3836	5251	5686	3799	5359	6140	2122	3266	3177
10	2513	4260	4961	5833	7492	8390	5179	7657	8232	3167	4094	5120
50	1179	5028	6575	16376	20672	22595	10718	15213	18063	6838	10487	11107
100	-2697	2688	4188	26962	33156	35557	14811	22026	24853	9455	13818	14892

Table 3: Value at Risk and Diversification Benefit across horizons – Monte Carlo with N=10000 realizations

5 Historical Simulation

5.1 Methodology

Historical simulation estimates VaR by revaluing the current portfolio under past market scenarios, without imposing any parametric distribution. This is particularly relevant given the deviations from normality observed in Figure 1.

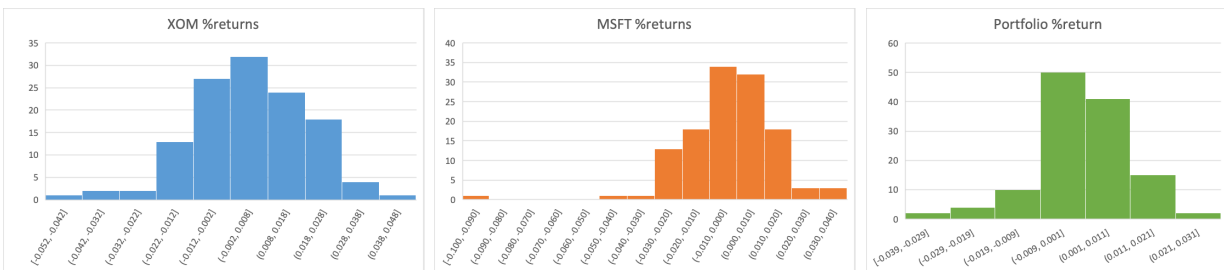

Figure 1: Return distributions of single assets and portfolio

A six-month window of daily returns is used to generate historical scenarios. Each return is applied to current prices as:

$$P_{\text{new}} = P_{\text{today}} \cdot \frac{P_i}{P_{i-1}} \tag{6}$$

The portfolio is revalued under each scenario, producing an empirical P&L distribution. One-day VaR is computed directly as the corresponding quantile.

For longer horizons, VaR is scaled using the square-root-of-time rule to maintain comparability with previous methods. This relies on assumptions of independence and constant volatility, which are not fully consistent with the historical framework. A fully consistent alternative would be multi-day historical aggregation, although this reduces the effective sample size and weakens tail estimates.

5.2 Results and Discussion

Results are broadly consistent with parametric approaches, despite the absence of distributional assumptions. Portfolio VaR remains lower than the sum of individual VaRs, confirming

	XOM VaR ($)			MSFT VaR ($)			Portfolio VaR ($)			Div. benefit ($)		
T	95%	99%	99.5%	95%	99%	99.5%	95%	99%	99.5%	95%	99%	99.5%
1	1072	1838	2179	1385	2340	3542	1842	3009	3444	616	1169	2277
5	2398	4110	4872	3098	5231	7920	4118	6728	7701	1377	2614	5091
10	3391	5812	6890	4381	7398	11201	5824	9514	10891	1948	3696	7200
50	7583	12997	15407	9796	16543	25047	13023	21275	24354	4356	8265	16100
100	10724	18381	21789	13854	23395	35421	18418	30087	34441	6160	11689	22769

Table 4: Value at Risk and Diversification Benefit across horizons – Historical simulation with 123 scenarios (6 months daily data)

diversification effects. However, the method is purely backward-looking and may not capture future market conditions not present in the historical sample.

6 Historical Simulation with Bootstrap

6.1 Methodology

The bootstrap approach improves on standard historical simulation by resampling returns with replacement from the empirical dataset, generating a larger number of scenarios.

For a two-asset portfolio, joint returns are resampled to preserve the dependence structure. At each step, an observed pair of returns is drawn and used to construct portfolio returns:

$$r_{p,t} = w_{1,t}\, r_{1,t} + w_{2,t}\, r_{2,t} \tag{7}$$

Multi-period returns are obtained by compounding:

$$R_{p,T} = \prod_{t=1}^{T} (1 + r_{p,t}) - 1 \tag{8}$$

This process is repeated N times to obtain the simulated P&L distribution, from which VaR is computed as the empirical quantile.

6.2 Results and Discussion

The bootstrap approach avoids the square-root-of-time scaling by directly generating multi-period outcomes, while preserving empirical features such as skewness and fat tails.

Compared to standard historical simulation, it provides more stable tail estimates by increasing the effective number of scenarios and preserving cross-asset dependence.

However, it still relies on the assumption that returns are i.i.d., and therefore does not capture time-varying volatility or structural market changes.

	XOM VaR ($)			MSFT VaR ($)			Portfolio VaR ($)			Div. benefit ($)		
T	95%	99%	99.5%	95%	99%	99.5%	95%	99%	99.5%	95%	99%	99.5%
1	1090	1859	2689	1387	2545	5134	1916	3071	4040	561	1334	3783
5	2149	3350	4088	4026	6491	7436	3840	5916	6677	2336	3924	4846
10	2533	4153	4821	5962	8407	9515	5526	7621	8872	2969	4940	5463
50	1402	5239	6839	15030	18253	19814	11030	15997	17439	5402	7494	9214
100	-2604	3323	4874	22028	25151	26871	14194	20532	23017	5230	7943	8727

Table 5: Value at Risk and Diversification Benefit across horizons – Historical simulation with Bootstrap

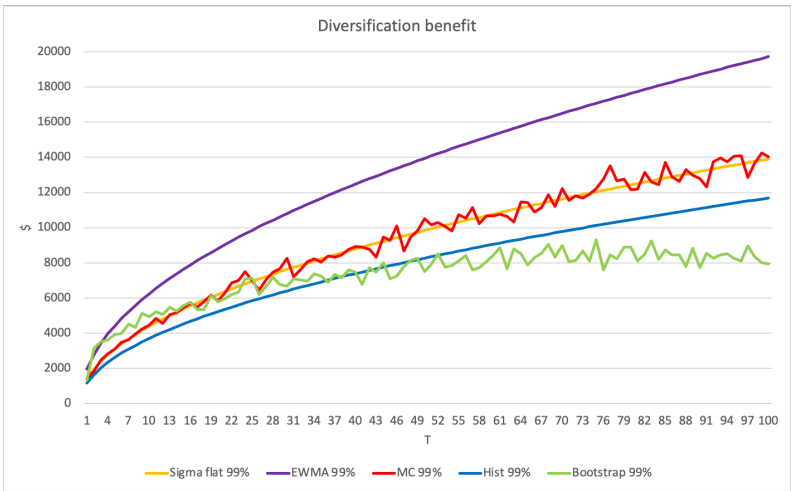

Figure 2: Diversification benefit across methods at fixed confidence level (99%)

7 Conclusions

Figure 2 shows that the diversification benefit increases with the time horizon for all approaches based on analytical scaling, confirming that imperfect correlation reduces portfolio risk more significantly over longer periods.

Monte Carlo results closely follow the parametric approach, as both rely on the assumption of normally distributed returns and square-root-of-time scaling. The small deviations observed are due to simulation noise.

The EWMA method produces the highest diversification benefit, reflecting higher volatility estimates driven by recent market conditions. In contrast, historical simulation yields lower values, as it is entirely based on past observations and does not incorporate forward-looking volatility dynamics.

The bootstrap approach differs from all other methods, as it does not rely on square-root-of-time scaling. Instead, multi-period returns are obtained through compounding, resulting in a flatter diversification benefit profile at longer horizons.

Overall, while all methods confirm the presence of diversification effects, the magnitude and growth over time depend critically on the assumptions regarding volatility dynamics and time aggregation.